23 September 2019 **STA2020F TUT 9** **NON-PARAMETRICS**

QUESTION 1

Polygraphs (lie detectors) are increasingly being used in many different areas of work. These machines depend on measuring small changes in the skin potential as a subject is asked a series of questions. The data below come from a bigger study in which independent groups of people were measured when feeling emotions of fear, happiness, depression and calmness.

Fear:	23.1	57.6	10.5	
Happiness:	22.7	53.2	9.7	19.6
Depression:	22.5	53.7	31.0	
Calmness:	14.8	13.3		

a) Use a suitable nonparametric test to investigate possible differences in location between these four groups. Conduct the test at a 1% significance level, and use the p-value approach.
b) Does the above data meet the requirements necessary to assume that the test statistic follows a chi-squared distribution? Comment on the implications of your answer with respect to the validity of the test

QUESTION 2

Each year the employees of a large firm are assessed for performance on a 7-point scale where 1 = very unsatisfactory and 7 = excellent. Management believes the assessment scores this year are related to last year's. To test this belief a random sample of 10 employees' scores from last year is drawn and the same employees' scores from this year are recorded. Do the data below support (at the 5% level) the management's belief? Identify (with reasons) and conduct a suitable statistical test, using a critical value.

This year:	5	6	4	5	5	4	7	6	4	5
Last year:	5	5	3	3	4	3	4	2	6	1

QUESTION 3

A well-known soft drink manufacturer has used the same secret recipe for its product since its introduction more than 100 years ago. In response to decreasing market share, however, the president of the company is contemplating changing the recipe. She has developed two alternative recipes. In a preliminary study, she asked 15 randomly selected people to taste the original recipe and the two new recipes. Each person was then asked to evaluate the product on a 5-point scale, where 1=awful, 2=poor, 3=fair, 4=good and 5=wonderful. The data is shown below:

Original	New Recipe 1	New Recipe 2
5	5	5
3	4	5
4	5	5
2	4	4
3	3	5
2	2	3
3	3	2
4	3	5
1	1	1
1	3	2
2	4	3
3	3	4
5	3	4
3	2	3
4	3	5

Can we conclude at the 1% significance level that there are differences in the ratings of the three recipes? Identify (with reasons) and conduct a suitable test.